

HIGH ACRYLONITRILE NBR: ORGANIC-SOLVENT FREE-RADICAL COPOLYMERIZATION AND COMPOSITION-CONTROLLED NITRILE RUBBER SYNTHESIS

1. ABSTRACT

This report reviews patent evidence relevant to high acrylonitrile nitrile butadiene rubber (NBR), with emphasis on synthesis routes that control acrylonitrile incorporation through monomer feed, solvent choice, initiator/regulator selection, and conversion management. The single primary patent in the manifest, EP2478021B1, discloses solution-phase free-radical preparation of nitrile rubbers in organic solvents using a conjugated diene, an α,β -unsaturated nitrile, and optional additional copolymerizable monomers, together with a modifier/regulator system. The patent is not limited to a single high-AN grade, but its examples explicitly demonstrate acrylonitrile/butadiene copolymerization in dimethylacetamide (DMAc) with controlled radical chemistry, as well as terpolymer variants containing hydroxyethyl acrylate, HDODA, and styrene. The disclosed examples show practical composition control via monomer charge ratios expressed in parts by weight per 100 parts monomer mixture, with acrylonitrile and butadiene fed at 37 phm and 63 phm in Example 1, and polymerization initiated at 100°C under 3 bar butadiene pressure. The process also uses oxygen exclusion, argon degassing, and post-polymerization precipitation in ethanol containing stabilizer. Across the examples, the patent emphasizes that solution polymerization can be used to prepare nitrile rubbers and functionalized nitrile rubbers with tailored molecular weight and architecture, including low-molecular-weight RAFT macromers and block/terpolymer structures. For high acrylonitrile NBR synthesis, the key technical message is that composition is governed primarily by monomer ratio and controlled radical conditions rather than by post-modification. The evidence supports solution polymerization in organic solvent, controlled initiator/regulator dosing, and conversion monitoring as the main process levers for achieving nitrile rubbers with targeted acrylonitrile content and manageable viscosity.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2478021B1
- Patent Title: Nitrile rubbers and production of same in organic solvents
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses free-radical solution polymerization of at least one conjugated diene and at least one α,β -unsaturated nitrile in an organic solvent in the presence of a modifier/regulator, with optional subsequent hydrogenation. The examples use DMAc as solvent, argon degassing/oxygen exclusion, and precipitation of the resulting nitrile rubber from ethanol containing stabilizer. - Acrylonitrile: 43 ml (653.3 mmol, 37 phm) — Example 1 - 1,3-Butadiene: 90 ml (1077 mmol, 63 phm) — Example 1 - Vazo® 88: 284.9 mg (1.17 mmol, 0.31 phm) — Example 1 - DoPAT: 204 mg (0.583 mmol, 0.22 phm) — Example 1 - Dimethylacetamide (DMAc): 50 ml (51 phm) — Example 1 - Argon degassing: 10 minutes — Example 1 - Butadiene feed pressure: 3 bar — Example 1 - Polymerization temperature: 100°C — Example 1 - Polymerization time: 9 hours — Example 1 - Final conversion: 39% — Example 6 - Regulator amount: 2.22 phm — Example 7 - Polymerization time: 9 h — Example 7 - Final conversion: 48% — Example 7 - Styrene: 2.79 g (26.8 mmol, 50 phm) — Example 8 - DMAc: 2 g — Example 8 - Polymerization temperature: 100°C — Example 8 - Polymerization time: 13 hours — Example 8 - Final conversion: 9.2% — Example 8 - Vazo® 88: 128 mg (0.523 mmol, 0.15 phm) — Examples 18, 20 and 23 - Post-dosing time: 2.5 and 6 hours — Examples 18, 20 and 23 - Butadiene feed pressure: 3 bar — Examples 17-23 - Polymerization temperature: 100°C — Examples 17-23 - Polymerization time: 9 hours — Examples 17-23

Relevant Experimental Evidence - Example 1: Demonstrated solution copolymerization of acrylonitrile and butadiene in DMAc at 100°C and 3 bar, yielding nitrile rubber under controlled radical conditions. - Example 6: Demonstrated terpolymer preparation by adding hydroxyethyl acrylate (HEA) with acrylonitrile; reaction stopped after 6 hours at 39% conversion. - Example 7: Demonstrated preparation of a low-molecular-weight NBR by increasing regulator amount to 2.22 phm and stopping at 48% conversion after 9 h. - Example 8: Demonstrated chain extension/block copolymer formation by reacting NBR macromer with styrene at 100°C in DMAc. - Example 17: Demonstrated broader NBR terpolymer recipes using acrylonitrile with optional HDODA/HEA under the same 100°C, 3 bar solution-polymerization platform.

Technical Relevance This patent is directly relevant because it provides an organic-solvent free-radical route to nitrile rubbers with explicit acrylonitrile/butadiene feed ratios and controlled radical conditions. Although not limited to a single high-AN grade, it shows the process levers most important for high acrylonitrile NBR: monomer ratio, solvent-based solution polymerization, regulator dosing, and conversion control.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

EP2478021B1 supports solution-phase free-radical synthesis as a viable route to nitrile rubbers with controlled composition and molecular weight. For high acrylonitrile NBR, the most evidence-based parameters are acrylonitrile/butadiene feed ratio control, oxygen-free operation, organic solvent use (DMAc), polymerization at 100°C, and controlled radical initiation/regulation with Vazo® 88 and DoPAT. The patent suggests that composition tuning is achieved during polymerization rather than by downstream modification, making it relevant for designing high-AN NBR recipes with manageable viscosity and tailored architecture.

5. REFERENCES

- EP2478021B1 | Nitrile rubbers and production of same in organic solvents | Not disclosed | EP | 2015 | <https://patents.google.com/patent/EP2478021B1/en>